LUND UNIVERSITY

Centre for Mathematical Sciences
*Division of Mathematics*

# Group and Ring Theory, Autumn Semester 2025
**Programme 2.9 – 28.10**

Starting on the 2nd of September 2025, the lectures will be on Tuesdays and Fridays 15:15 – 17:00 in room MH:228. Starting on the 11th of September 2025, the exercise classes will be on Thursdays at 15:15 – 16:00 in MH:228. The lecture notes and exercise sheets will be uploaded on Canvas in good time.

**Optional Literature:**
[B] P. B. Bhattacharya, S. K. Jain and S. R. Nagpaul, *Basic Abstract Algebra,* 2nd ed., Cambridge University Press, 1994.

**Tuesday 2/9** Study: Group actions (e.g. Section 5.4 to 5.4.9 in [B])
Solve: Exercise Sheet 1

**Friday 5/9** Study: Applications of the Orbit-Stabiliser (e.g. Section 5.4.10 to 5.4.13 in [B])
Solve: Exercise Sheet 2

**Tuesday 9/9** Study: Sylow theorems (e.g. Section 8.4 to 8.4.6 in [B])
Solve: Exercise Sheet 3

**Thursday 11/9** Discuss: Exercise Sheets 1 and 2 at exercise class

**Friday 12/9** Study: Applications of Sylow theorems (e.g. Section 8.4.7 to 8.5 in [B])
Solve: Exercise Sheet 4

**Tuesday 16/9** Study: An introduction to modules (e.g. Section 14.1 to 14.3 in [B])
Solve: Exercise Sheet 5

**Thursday 18/9** Discuss: Exercise Sheets 3 and 4 at exercise class

**Friday 19/9** Study: Completely reducible modules (e.g. Section 14.4 in [B])
Solve: Exercise Sheet 6

**Tuesday 23/9** Study: Free modules (e.g. Section 14.5 in [B])
Solve: Exercise Sheet 7

**Thursday 25/9** Discuss: Exercise Sheets 5 and 6 at exercise class

**Friday 26/9** Study: Noetherian and Artinian modules (e.g. Section 19.1 to 19.2.4 in [B])
Solve: Exercise Sheet 8

**Tuesday 30/9** Study: Noetherian and Artinian rings (e.g. Section 19.2.5 to 19.2.12 in [B])
Solve: Exercise Sheet 9

**Thursday 2/10** Discuss: Exercise Sheets 7 and 8 at exercise class

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| **Friday 3/10** | Study: Hilbert basis theorem (e.g. Section 19.2.14 to 19.2.15 in [B])<br>Solve: Exercise Sheet 10 |
| **Tuesday 7/10** | Study: Wedderburn-Artin Theorem (e.g. Section 19.3 to 19.3.6 in [B])<br>Solve: Exercise Sheet 11 |
| **Thursday 9/10** | Discuss: Exercise Sheets 9 and 10 at exercise class |
| **Friday 10/10** | Study: Smith normal form over a PID (e.g. Chapter 20 in [B])<br>Solve: Exercise Sheet 12 |
| **Tuesday 14/10** | Study: Finitely generated modules over a PID (e.g. Section 21.1 to 21.3 in [B])<br>Solve: Exercise Sheet 13 |
| **Thursday 16/10** | Discuss: Exercise Sheets 11 and 12 at exercise class |
| **Friday 17/10** | Study: Rational canonical form and tensor products (e.g. Sections 21.4 and 22.2 in [B])<br>Solve: Exercise Sheet 14 |
| **Tuesday 21/10** | Study: Module structure of the tensor product (e.g. Section 22.3 to 22.4 in [B])<br>Solve: Exercise Sheet 15 |
| **Thursday 23/10** | Discuss: Exercise Sheets 13, 14 and 15 at exercise class |
| **Friday 24/10** | Revision session 1<br>Discuss: The written exams from August 2022 and August 2023. |
| **Tuesday 28/10** | Revision session 2<br>Discuss: The written exams from November 2024 and August 2025. |

The written exam will take place on the 31st of October 2025 at 8:00–13:00. The associated oral exams will take place in the afternoons of the week afterwards.

The retake written exam will take place on the 29th of November 2025 at 8:00–13:00. The associated oral exams will take place in the afternoons of the week afterwards.

More information and instructions for the oral exams will be posted on the Canvas page in due course.